Saqlain Afroz

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About

PhD student at IFAE Barcelona, working on gravitational wave physics and machine learning. My research focuses on fast parameter estimation, signal classification, waveform modelling, detector characterization. Interested in general relativity, gravitational wave astrophysics, machine learning, quantum computation, and multi-messenger astronomy.

Education & Positions

Indian Institute of Science Education and Research Kolkata *December 2020 – May 2025*
BS-MS Dual Degree in Physical Sciences

- **Some Relevant Coursework:** Introductory Astrophysics, General Theory of Relativity and Cosmology, Fluid and Magnetohydrodynamics, Statistics and Probability, Computational Astrophysics, Topology, Computer Science

Inter-University Center for Astronomy and Astrophysics, Pune *August 2024 – April 2025*
*Master's Thesis (**Supervisor:** Dr. Apratim Ganguly)*

- Developed a deep learning pipeline leveraging **normalizing flows** for rapid parameter estimation of gravitationally lensed (microlensed) gravitational waves, considering Gaussian as well as real noise, offering a significant speed-up over traditional Bayesian inference methods like bilby, with inference time being $\mathcal{O}(sec)$.
- Designed and trained **Convolutional Neural Networks** and **Vision Transformers** for classifying gravitational wave signals into distinct physical categories: lensed, precessing, eccentric, and unlensed, with high accuracy, $\gtrsim 90\%$.

International Centre for Theoretical Sciences, Bangalore *October 2025 – November 2025*
*Visiting Fellow (**Supervisor:** Prof. Prayush Kumar)*

- Worked on my project of developing a surrogate model of eccentric binaries by efficiently extracting more eccentric harmonics.

Institut de Física d'Altes Energies, Barcelona, Spain *September 2026 – Ongoing*
*PhD Student (**Supervisor:** Prof. Andrew Lundgren, Prof. Mario Martinez)*

- Working primarily on GW searches and parameter estimation, alongwith detector characterization and instrument baffles.

Research Experiences

Surrogate Modelling of Gravitational Waves (ICTS-TIFR) *September 2025 - August 2026*

Guide: Prof. Prayush Kumar

- Using the Reduced Order Method and Interpolation Method to build surrogate models of different types of gravitational waves.

Quasi-Normal Mode Computation of Black Holes via Physics-Informed Neural Networks (IUCAA) *February 2025 - March 2026*

Guide: Prof. Ajit Kembhavi, Prof. Dawood Kothawala, Dr. Ninan Sajeeth Philip

- Developed a Physics-Informed Neural Network (PINN) framework to compute quasi-normal modes of black holes in alternative theories of gravity and black holes in dark matter profiles.

Classification of Blazars using ANNs (BHU) *April 2025 – October 2025*

Advisor: Prof. Raj Prince

- Developed an Artificial Neural Network (ANN) that learns to classify Blazars into BL Lac and FSRQ.

Numerical Relativity (IISER-Kolkata) *August 2024 – December 2024*

Advisor: Prof. Rajesh Kumble Nayak

- Studied the $3+1$ ADM decomposition of Einstein's field equations for numerical evolution of spacetime metrics.

Quantum Simulation of Hawking Radiation using VQE on IBM Quantum Computer (IISER-Kolkata) *December 2022 – August 2023*

Guide: Prof. Prasanta K. Panigrahi

- Designed and implemented a quantum simulation framework to study Hawking radiation using the Variational Quantum Eigensolver (VQE) algorithm to model black hole-inspired Hamiltonians and simulate near-horizon quantum effects.

Natural Language Processing Project (IISER-Kolkata) *August 2022 - January 2023*

Guide: Prof. Kripabandhu Ghosh

- Developed a machine learning model that can do segmentation on clauses from Law contracts and then do clause classification.

Skills

Programming Languages: Python, Rust, CUDA, Julia, Matlab, Arduino, C++, C, Linux, LaTeX, JavaScript, NextJS
High Performance Computing: HTCondor, SLURM, PBS, OpenMP, MPI
Deep Learning & Machine Learning: PyTorch, Tensorflow, normflows, nflows, CNNs, GANs, GNNs, RNNs, LSTMs, SVMs, Transformers, Autoencoders, Physics Informed Neural Network, Decision Trees, K-means, k-NN, Random Forests
Statistical Techniques: Bayesian Inference, Maximum Likelihood Estimation, Hypothesis Testing, Markov Chain Monte Carlo (MCMC), Regression Analysis, Time Series Analysis, Singular Value Decomposition, Surrogate, Gaussian Process Regression, Greedy Algorithm
Software Tools & Frameworks: Git, Matlab, Colab, Docker, PyCBC, GWpy, Bilby, Dingo, Qiskit, NumPy, SciPy etc
Spoken Languages: English (proficient), Español (intermediate), Hindi(Native)

Invited Talks & Presentations

Normalizing Flows in Gravitational Wave Parameter Estimation, IISER-K, 2025
Presented my work on using Normalizing flows, a generative model, to do parameter estimation of lensed gravitational waves and motivated the students to adopt ML approaches in their own study.

Gravitational Wave Open Data Workshop, 2025
Appointed as an instructor to mentor and teach gravitational wave astronomy at the GW Open Data Workshop's Study Hub held at IISER Kolkata. I explained the theory of GWs, their data analysis techniques, and some facts about instrumentation.

ML4HEP Workshop, 2025
Recognized for expertise in ML, I was invited to teach at this national level workshop hosted by IISER Kolkata in June 2025.

National Science Day Seminar, 2025
Presented a public lecture at IUCAA on National Science Day, reflecting on how modern physics, through General Relativity and Quantum Mechanics, challenges our everyday notions of reality.

Workshops and Conferences Attended

The Future of Gravitational Wave Astronomy, ICTS-TIFR, 2025
AI / ML Workshop at IUCAA, 2025 *Cert.*
LIGO Workshop, BITS Pilani and IUCAA, 2024 *Cert.*
NVIDIA AI Summit, Mumbai, 2024 *Cert.*
Classical and Quantum Gravity Conference, Cochin University, 2024 *Cert.*
GWOSC Workshop 5, 2022 *Cert.*
Quantum Information and Quantum Technology *Cert.*
Qiskit Global Summer School 2022 *Cert.*

Awards

Duolingo English Test *Cert.*
Achieved a perfect score of 150/160.

UP Science Talent Search Examination Scholarship *Cert.*
A scholarship awarded by the State government for students highly talented in Basic Sciences.

NCC Special Trophy *Cert.*
Was awarded a special trophy from the National Cadet Corps, of the Indian Armed Forces, for my excellent academic performance.

Robotics
Built a semi-autonomous robot using Arduino that can detect obstacles in its path and safely tackle them, using sensors.

Publications

- **S. Afroz**; A. Ganguly, "Novel Approach in Deep Learning Based Classification of Gravitational Waves" (Completed)
- **S. Afroz**; A. Ganguly, "Deep Learning Based Parameter Estimation of Lensed Gravitational Waves" (Under preparation)
- **S. Afroz**; T. Mukherjee; R. Prince, "Using ML to classify Blazars using multi-wavelength data" (Submitted on arXiv here)

Extracurricular Activities

- **Chess:** Secured top-10 position in multiple chess tournaments at IISER Kolkata as well as IUCAA.
- **Leadership & Management:** Served as **Secretary of IISER Kolkata's Mess** for two consecutive years and held the position of **Office Bearer of the Institute's Gym**. Also managed the **official webpage of the Department of Physics, IISER Kolkata**.